Snake Game AI agent using Rule-Based Agent

Omika Maurya, Ayan Ahamad, Anurag Yamuna(2CSE13)
IILM University Greater Noida

Abstract

This paper presents the design and implementation of an AI-based Snake Game developed using Python and Pygame. The system utilizes a rule-based intelligent agent to control the movement of the snake automatically. The agent makes decisions based on the position of the food and ensures safe movement by avoiding collisions with walls and its own body. The project demonstrates the application of artificial intelligence concepts in game development. Although reinforcement learning techniques were explored during the research phase, the final implementation is based on rule-based logic due to simplicity and time constraints. The system successfully demonstrates automated gameplay with efficient decision-making.

Keywords: Artificial Intelligence, Snake Game, Rule-Based Agent, Pygame

1. Introduction

Artificial Intelligence (AI) has become one of the most important fields in computer science, enabling systems to perform tasks that typically require human intelligence. AI techniques are widely used in areas such as robotics, gaming, healthcare, and automation. One of the simplest and most effective ways to understand AI concepts is through game development.

The Snake Game is a classic arcade game that provides a controlled environment for implementing AI algorithms. The game involves a snake that moves in a grid-based environment to collect food while avoiding collisions with walls and itself. As the snake grows longer, the complexity of the game increases, making it a suitable problem for studying decision-making and path planning.

In this project, an AI-based approach is used where an intelligent agent controls the snake automatically. The agent continuously observes the environment and makes decisions to maximize the score by collecting food while avoiding collisions. This project demonstrates the practical application of AI concepts in a simple and interactive environment.

2. Objectives

The primary objective of this project is to design and develop an AI-based Snake Game that demonstrates the application of artificial intelligence techniques in a controlled gaming environment. The project aims to provide a clear understanding of how intelligent agents can be used to automate decision-making processes in real-time systems.

The specific objectives of the project are as follows:

- To design and implement a fully functional Snake Game using Python programming language and the Pygame library.
- To develop an intelligent agent capable of controlling the snake automatically without human intervention.
- To enable the agent to analyze the game environment, including the position of the snake, food, and boundaries, in order to make effective decisions.
- To implement rule-based decision-making logic that allows the snake to

move toward the food efficiently while avoiding collisions with walls and its own body.

- To incorporate safety mechanisms that ensure the snake selects only valid and non-colliding paths during gameplay.
- To study and understand fundamental artificial intelligence concepts such as intelligent agents, decision-making, and environment interaction.
- To explore reinforcement learning techniques such as Q-learning conceptually and understand their potential application in game environments.
- To evaluate the performance of the AI agent in terms of its ability to survive longer and achieve higher scores.
- To identify the limitations of rule-based systems and highlight the need for more advanced AI approaches.
- To provide a foundation for future enhancements, including the integration of reinforcement learning, pathfinding algorithms, and machine learning techniques.

---

3. Methodology

The methodology of the project is based on the implementation of a rule-based intelligent agent. The system operates in a grid-based environment where the snake moves in discrete steps. The agent continuously monitors the state of the environment, which includes the position of the snake, the location of the food, and the boundaries of the game window.

The decision-making process of the agent involves the following steps:

1. **State Observation:**
   The agent collects information about the current position of the snake and the food.
2. **Direction Analysis:**
   The agent compares the coordinates of the snake and the food to determine the optimal direction of movement.
3. **Safety Check:**
   Before making a move, the agent checks whether the next position will result in a collision with the wall or the snake's body.
4. **Action Selection:**
   The agent selects a safe movement direction (up, down, left, or right) based on predefined rules.
5. **Fallback Strategy:**
   If the direct path to the food is unsafe, the agent selects an alternative safe path randomly.

Although reinforcement learning techniques such as Q-learning and Deep Q-Networks (DQN) were studied during the research phase, they were not implemented in the final system. Instead, a rule-based approach was chosen due to its simplicity, faster execution, and ease of understanding.

---

4. System Architecture

The system architecture consists of multiple components working together to achieve automated gameplay.

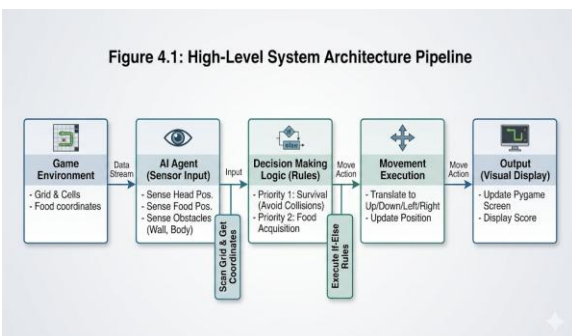
Components:

- **Game Environment:** Displays the game interface and manages rendering
- **AI Agent:** Responsible for decision-making
- **Snake Module:** Handles movement and growth
- **Food Module:** Generates food at random positions
- **Collision Detection Module:** Detects game-over conditions

Workflow:

Game Input → AI Agent → Decision Logic → Movement Execution → Game Output

The AI agent acts as the core component that connects all modules and controls the behavior of the snake.



The system is divided into two main modules:

1. Game Module

- Handles rendering of the snake and food
- Manages score tracking
- Detects collisions
- Controls game speed

2. Agent Module

- Implements decision-making logic
- Determines the next movement direction
- Ensures safe navigation

The agent uses conditional statements to evaluate different scenarios and select the best possible action. The implementation is efficient and runs smoothly in real-time.

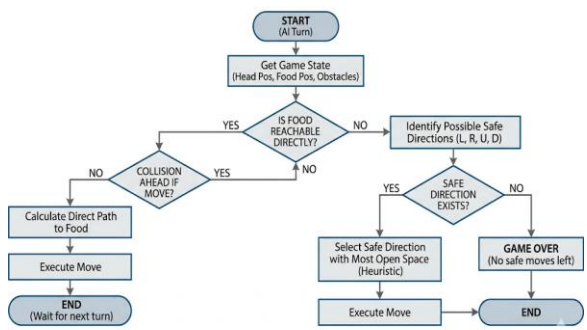


Figure 5.1: Rule-Based AI Agent Decision Flowchart

5. Implementation

The implementation of the project is carried out using Python programming language along with the Pygame library. Pygame provides functionalities for creating graphical interfaces, handling events, and rendering objects on the screen.

6. Results and Discussion

The developed system successfully demonstrates the use of an AI agent in controlling the snake automatically. The agent is capable of:

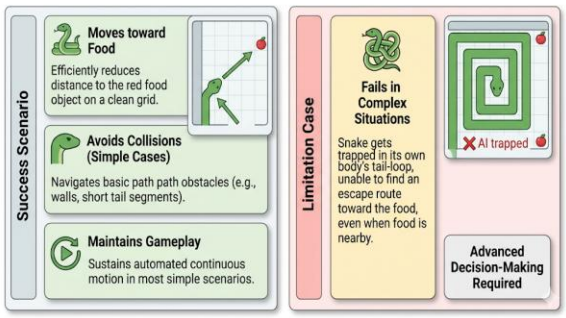
- Moving towards the food efficiently
- Avoiding collisions with walls

- Preventing self-collision in most cases
- Maintaining continuous gameplay

The performance of the system is satisfactory for a rule-based approach. However, in complex situations where multiple paths are available, the agent may not always choose the optimal path.

The results show that even simple AI techniques can produce effective outcomes in controlled environments.

Figure 6.1: Results and Discussion Analysis - Rule-Based AI Success vs. Limitation



---

7. Limitations

The project has the following limitations:

- Uses basic rule-based logic
- No learning capability
- Performance is limited in complex scenarios

---

8. Future Scope

The project can be extended and improved in several ways:

- Implementation of **Reinforcement Learning (Q-Learning, DQN)**
- Use of **Pathfinding Algorithms (BFS, A*)**
- Development of a **Web-based version using JavaScript**
- Integration of **Machine Learning models for adaptive behavior**
- Improvement in decision-making using **heuristic techniques**

These enhancements can significantly improve the intelligence and performance of the system.

---

9. Conclusion

This project presents a detailed implementation of an AI-based Snake Game using a rule-based intelligent agent. The system demonstrates how artificial intelligence concepts can be applied in simple game environments to automate decision-making processes.

The project successfully achieves its objectives by enabling the snake to move autonomously, collect food, and avoid collisions. Although the current implementation is based on rule-based logic, it provides a strong foundation for future enhancements using advanced AI techniques such as reinforcement learning.

Overall, the project highlights the importance of AI in modern applications and its potential to solve complex problems in an efficient manner.

---

10. References

[1] Stuart Russell and Peter Norvig, *Artificial Intelligence: A Modern Approach*, 2010.

[2] Richard S. Sutton and Andrew G. Barto, *Reinforcement Learning: An Introduction*, 2018.

[3] Python Documentation, Available online: https://docs.python.org

[4] Pygame Documentation, Available online: https://www.pygame.org/docs

[5] Volodymyr Mnih et al., "Playing Atari with Deep Reinforcement Learning," 2013.

[6] David Silver et al., "Mastering the Game of Go with Deep Neural Networks," 2016.

[7] John McCarthy, "What is Artificial Intelligence?", 2007.